

Mahija Ibad Pradipta

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EDUCATION

Sepuluh Nopember Institute of Technology — Surabaya

August 2022 – Present

Bachelor of Computer Engineering

GPA: 3.21/4.00

SMAN 4 Jakarta — Jakarta

July 2019 – May 2022

Science Major

Final Score: 86/100

WORK EXPERIENCE

Digital Signal Processing Laboratory Course — M-IOT ITS

October 2025 – December 2025

Staff - Teaching Assistant

On-site

- Assisted and supervised students during Digital Signal Processing laboratory sessions
- Evaluated students' understanding of course materials, assignment completion, and discipline throughout laboratory activities
- Reviewed and assessed Digital Signal Processing laboratory reports

Introduction to Programming Laboratory Course — M-IOT ITS

July 2025 – December 2025

Secretary and Treasurer - Teaching Assistant

On-site

- Supervised, guided, and evaluated students during laboratory sessions, mentoring activities, laboratory reports, and final projects
- Prepared laboratory modules and managed administrative activities
- Managed attendance records and assessment recapitulation for teaching assistants and students

PT Winnicode Garuda Indonesia — Bandung

March 2025 – September 2025

Intern - Full-Stack Developer

Remote

- Developed UNTERN (U Need Intern), a web application connecting internship seekers with companies seeking talent
- Managed the entire application development lifecycle, including design, implementation, testing, and deployment
- Analyzed user requirements and translated them into application features and functionalities

PT Lintasarta — Central Jakarta

July 2025 – August 2025

Intern - IT Services Management (Infrastructure)

On-site

- Supported activities within the IT Infrastructure subdivision under the IT Services Management Department, Corporate IT Division
- Participated in hardware deployment processes to support company operations
- Learned enterprise network architecture and IT service management practices within a corporate environment

Introduction to Programming Laboratory Course — M-IOT ITS

September 2024 – March 2025

Staff - Teaching Assistant

On-site

- Assisted Electrical Engineering, Telecommunications Engineering, and Biomedical Engineering students during Introduction to Programming laboratory sessions
- Guided students through Introduction to Programming mentoring activities
- Evaluated final project presentations and laboratory reports

Robotics Extracurricular — SDN Klampis Ngasem I/246 - Surabaya

August 2023 – June 2024

Instructor

On-site

- Guided and mentored 4th, 5th, and 6th grade students participating in robotics extracurricular activities
- Taught fundamental robotics concepts through theory and hands-on practice using the 4-Wheel Obstacle Avoidance Robot Kit
- Delivered robotics learning materials through interactive web-based simulations

PROJECTS

- DIY Smart CCTV Monitoring System** | *Docker, EMQX, MinIO, MySQL, Laravel, Webmin, Cloudflared* 2026
- Architected and deployed a self-hosted CCTV platform on an Intel NUC server integrated with ESP32-CAM edge devices for image collection and monitoring
 - Managed a multi-service Docker environment including Laravel, MySQL, MinIO, phpMyAdmin, EMQX MQTT broker, and supporting application services
 - Implemented MQTT-based data communication, server administration and monitoring through Webmin, and secure public access using Cloudflare Tunnel
- Autonomous Human Search System Using Drone with Pi 5** | *YOLO, ONNX, MAVSDK, Edge Device* 2026
- Designed and integrated a drone power distribution system capable of providing stable power for Raspberry Pi 5 computing requirements
 - Evaluated and optimized YOLOv8n ONNX model performance in a headless Linux environment on Raspberry Pi 5 for real-time human detection
 - Developed an autonomous MAVSDK-based system with autonomous navigation, victim documentation, and duplicate target tracking prevention capabilities
- Self-Hosted Homelab Infrastructure** | *Debian, Docker, Cloudflare Tunnel, Webmin, Linux Administration* 2025
- Built and maintained a self-hosted homelab environment using repurposed laptop hardware to run personal applications and services
 - Deployed and managed multiple web applications and supporting services using Docker containers on a Debian-based server
 - Implemented secure remote access through Cloudflare Tunnel and performed server administration and monitoring using Webmin
- SwiMate** | *Flutter, Dart, Supabase, PostgreSQL, TensorFlow Lite* 2025
- Developed a mobile application to support swimming training and performance monitoring
 - Integrated mobile sensors and machine learning models to support activity analysis
 - Implemented backend services and data management using Supabase and PostgreSQL
- UNTERN** | *JavaScript, PostgreSQL, Vite* 2025
- Developed a web platform connecting internship seekers with companies seeking talent
 - Managed the entire software development lifecycle from requirements analysis to deployment
 - Implemented application features and system functionalities to support the internship recruitment process
- SARVIO-X** | *JavaScript, DJI Tello, PyTorch* 2025
- Developed a web-based application for monitoring and interacting with DJI Tello drones
 - Implemented user interface components and drone control functionalities
 - Supported drone operations and monitoring through a browser-based platform
- Course Web Application** | *JavaScript, PostgreSQL* 2025
- Developed a web application for course management and learning activities
 - Implemented frontend and backend functionalities to support academic workflows
 - Designed features for managing course-related information and user interactions
- CarVole 2D Car Game** | *C++, graphics.h* 2024
- Developed a 2D car game using C++ and graphics.h
 - Implemented game mechanics, object movement, and collision handling
 - Designed interactive gameplay elements and user controls
- GUI Application with wxWidgets** | *C++, wxWidgets* 2023
- Developed a desktop graphical user interface application using wxWidgets
 - Applied object-oriented programming principles in application development
 - Implemented interactive user interface components and application features

AWARDS

With Banyubramanta ITS:

5th Place - Singapore Autonomous Underwater Vehicle Challenge (SAUVC) 2025	March 2025
1st Place - Indonesian Underwater Robot Competition National Level 2024	July 2024
3rd Place - Indonesian Underwater Robot Competition Regional Level II 2024	June 2024

TRAINING

Intermediate Student Management Skills Training (LKMM-TM)	August 2024
Basic Student Management Skills Training (LKMM-TD)	November 2023
Pre-Basic Student Management Skills Training	September 2023
MAGE 9 Multimedia Workshop	August 2023
Basic Student Entrepreneurship Skills Training (LKMW-TD)	November 2022
Basic Scientific Writing Training (PKTI-TD)	October 2022

ORGANIZATIONS

M-IOT Laboratory — Computer Engineering ITS — Surabaya	June 2025 – May 2026
<i>Coordinator - Board of Directors</i>	<i>On-site</i>
• Improved laboratory quality in both project development and human resource development aspects • Supervised overall timelines and task execution across all divisions and subdivisions • Coordinated laboratory management activities with external stakeholders and university administration	
HIMATEKKOM ITS — Surabaya	March 2025 – February 2026
<i>Head of Cadre Development Department - Board of Directors</i>	<i>On-site</i>
• Guided IC and OC teams in executing all cadre development activities • Mentored SC members in designing assessment systems and activity timelines • Coordinated with students and the Computer Engineering Department regarding cadre development activities	
M-IOT Laboratory — Computer Engineering ITS — Surabaya	August 2024 – June 2025
<i>Archive Staff - Internal Division</i>	<i>On-site</i>
• Managed archives of thesis topics and telematics projects conducted by M-IOT Laboratory assistants • Maintained archives of midterm and final examination materials for M-IOT specialization courses • Compiled learning materials for M-IOT specialization courses	
Banyubramanta Robotics Team ITS — Surabaya	July 2024 – May 2025
<i>Head of Official Division - Board of Directors</i>	<i>On-site</i>
• Led the Administration & Finance, Public Relations & Sponsorship, and Creative & Branding subdivisions • Planned and supervised overall timelines and task execution across all subdivisions • Ensured smooth execution of the team's administrative and non-technical operations	
HIMATEKKOM ITS — Surabaya	March 2024 – March 2025
<i>Secretary and Treasurer Staff - Board of Directors</i>	<i>On-site</i>
• Coordinated with the Computer Engineering Department regarding organizational administration • Prepared outgoing letters, meeting minutes, and certificate submissions for organizational activities • Maintained archives of important documents in both digital and physical formats	
HIMATEKKOM ITS — Surabaya	April 2024 – December 2024
<i>Steering Committee Staff - Cadre Development Department</i>	<i>On-site</i>
• Coordinated with IC (Instructor Committee) and OC (Organizing Committee) regarding cadre development activities • Designed student assessment systems and overall cadre development timelines • Coordinated with students and the Computer Engineering Department regarding cadre development activities	
Banyubramanta Robotics Team ITS — Surabaya	February 2023 – July 2024
<i>Senior Finance Staff - Official Division</i>	<i>On-site</i>
• Recorded team income and expenditures	

- Managed team budget allocation
- Prepared accountability reports, financial reports, and budget plans

Banyubramanta Robotics Team ITS — Surabaya

November 2022 – January 2023

Finance and Administration Intern - Official Division

On-site

- Prepared sponsorship and administrative proposals
- Prepared contracts, accountability reports, and budget plans
- Created content for the team's social media platforms

ITS Robotics Club — Surabaya

September 2022 – February 2023

Member

On-site

- Actively participated in organizational programs and activities

ITS IBC Club — Surabaya

September 2022 – September 2023

Member

On-site

- Actively participated in organizational programs and activities

COMMITTEES

Multimedia And Game Event 10 — Computer Engineering ITS

February 2024 – February 2025

Head of Event Division - Board of Directors

On-site

- Guided staff in designing and executing IoT, Multimedia, and Robotics workshops
- Guided staff in planning and executing opening ceremonies, talk shows, exhibitions, and closing events
- Supervised and evaluated the implementation of workshops, talk shows, exhibitions, opening ceremonies, and closing events

Multimedia And Game Event 9 — Computer Engineering ITS

February 2023 – March 2024

Staff - Secretariat Division

On-site

- Managed activity data collection and documentation
- Prepared communication materials, attendance records, feedback forms, and meeting minutes
- Served as the person in charge of one workshop

HGTS X SE 8.0 — HIMATEKKOM ITS

October 2023

Staff - Teaching Division

On-site

- Designed learning concepts for students of MTs 19 Surabaya
- Prepared presentation materials for teaching activities
- Taught image detection and pose detection topics to students of MTs 19 Surabaya

ISC - 63rd ITS Anniversary — ITS

September 2023 – November 2023

Badminton Staff - Event Division

On-site

- Ensured the smooth execution of the inter-faculty badminton tournament
- Coordinated match schedules and tournament facilities
- Managed documentation and match result records

TDC Summit Fest 2023 — TDC ITS

August 2023 – November 2023

TDC For Startup Staff - Event Division

On-site

- Prepared implementation and technical guidelines for startup bootcamp mentoring activities
- Prepared implementation and technical guidelines for startup bootcamp coaching activities
- Served as Master of Ceremony (MC) during startup bootcamp mentoring sessions

SPS 128 — HIMATEKKOM ITS

June 2023 – September 2023

Staff - Event Division

On-site

- Prepared implementation and technical guidelines for the 128th FTE ITS Graduation Appreciation Event
- Prepared event rundown for the 128th FTE ITS Graduation Appreciation Event
- Served as Liaison Officer for the Head of the Computer Engineering Department

Inclenation 2023 — BEM FTEIC ITS

June 2023 – August 2023

Staff - Mentor Division

On-site

- Mentored 20 freshmen from 6 departments and 10 study programs within FTEIC
- Provided guidance and support to freshmen throughout Inclenation 2023
- Served as an internalization facilitator during Inclenation 2023

SKILLS

- **Programming Languages:** JavaScript, Python, C++, Dart, SQL
- **Web Development:** Laravel, Vite, REST API, PostgreSQL, MySQL, React
- **Mobile Development:** Flutter
- **Infrastructure & DevOps:** Linux (Debian & Ubuntu), Docker, Cloudflare Tunnel, Webmin, MinIO, EMQX
- **Networking:** TCP/IP, MQTT, Network Troubleshooting, WebSocket
- **AI & Computer Vision:** YOLO, ONNX, TensorFlow Lite, OpenCV
- **Tools:** Git, GitHub, Postman, phpMyAdmin
- **Languages:** Indonesian (Native), Javanese (Conversational), English (Professional Working Proficiency)